



University of Management & Technology
Department of Computer Science
School of Systems and Technology
Fall Semester 2025

Instructor's Name:	<u>Syed Hamed Raza</u>	Subject:	<u>Discrete Structures</u>
Date:	19-11-2025	Subject Code:	
Quiz #:	<u>1</u>	Total Marks:	<u>20</u>
Student Name:		Reg #:	
Time Allowed:	<u>45 Minutes</u>		

Instructions for Candidates

Read the instructions carefully.

- i. Write Question number and its part number clearly.
 - ii. Attempt all questions.
 - iii. Cutting or over-writing is not acceptable
 - iv. Illustrate your answers with diagrams, graphs and tables if required.
 - v. Cross off blank pages before handing over the Answer-book to the examination staff at completion of the examination.
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Q1.

Let

$$S = \{3, 6, 9, 12, 15, 18\}.$$

- 1. Write S in set-builder notation using a predicate.**
- 2. Is S a subset of the set of even integers? Justify.**

Solution:

Q2.

Let

$$T = \{\mathbf{0}, \{\mathbf{0}\}\}.$$

- 1. List all elements of $P(T)$.**
- 2. Compute $|P(P(T))|$.**
- 3. State whether $\emptyset$ is an element of $P(T)$, a subset of $P(T)$, both, or neither.**

Solution:

Q3.

Given

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, A = \{1, 4, 5, 9\}, B = \{2, 4, 6, 8, 9\}, C = \{1, 2, 3, 9\}.$$

Compute:

- a) $(A - B) \cup (C - A)$**
- b) $A \Delta C$**
- c) $(A^c \cap B) \cup C^c$**

Solution:

Q4.

In a class of 80 students:

- 30 play Cricket (C)
 - 28 play Football (F)
 - 25 play Badminton (B)
 - 12 play both C and F
 - 9 play both F and B
 - 7 play both C and B
 - 4 play all three sports
1. Determine how many students play only Cricket.
 2. Determine how many students play exactly two sports.
 3. How many students play none of the sports?

Solution:

Q5.

A sequence is defined as

$$a_n = (-1)^n(3n - 1).$$

- 1. Compute a_0, a_1, a_2, a_3, a_4 .**
- 2. Identify the subsequence of all non-negative terms.**
- 3. Explain whether the absolute values $|a_n|$ form an arithmetic or geometric sequence.**

Solution:

Q6.

Consider the sequence:

$$5, 11, 23, 47, 95, \dots$$

- 1. Determine whether the sequence is arithmetic, geometric, or neither.**
- 2. Derive a recurrence relation for the sequence.**
- 3. Predict the next term using your recurrence.**

Solution:

Q7.

Evaluate:

$$\sum_{k=0}^n (5 - 2k).$$

Then:

- 1. Simplify the expression into a polynomial in n .**
- 2. Evaluate at $n = 40$.**
- 3. Explain why the summation represents the sum of a decreasing arithmetic sequence.**

Solution:

Q8.

A sequence satisfies:

$$a_n = 3a_{n-1} - 2a_{n-2}, a_0 = 4, a_1 = 9.$$

- 1. Compute a_2, a_3, a_4 .**
- 2. Solve the recurrence using the characteristic equation.**
- 3. Provide the explicit closed-form expression for a_n .**

Solution: